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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Machine Learning (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

- ☐ Introduction to Machine Learning (unit? unit=22&lesson=23)
- ☐ Supervised Learning (unit? unit=22&lesson=24)
- ☐ Unsupervised Learning (unit? unit=22&lesson=25)
- ☐ Reinforcement Learning (unit?)

Week 1: Assignment 1 (Non Graded)

Assignment not submitted

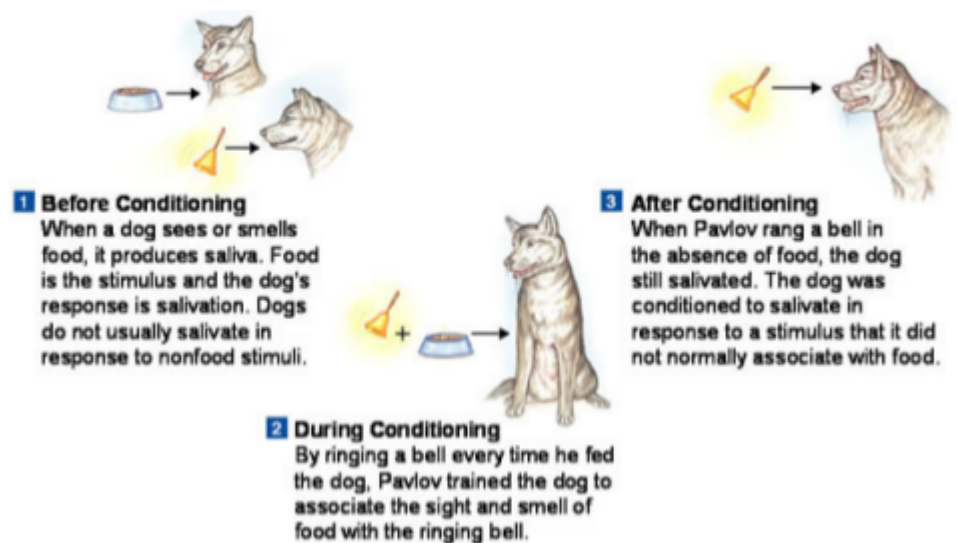
Note : This assignment is only for practice purpose and it will not be counted towards the Final score.

1) Which of the following tasks can be best solved using clustering.

1 point

- ☐ Predicting the amount of rainfall based on various cues
- ☐ Detecting fraudulent credit card transactions
- ☐ Training a robot to solve a maze

2) Pavlov's experiment is a classic experiment conducted by the Russian physiologist Ivan Pavlov. He experiments on dogs shown in the image below. **1 point**

(Image source - <http://www.goldiesroom.org/> (<http://www.goldiesroom.org/>))

unit=22&lesso
n=26)

☐ Statistical
Decision
Theory -
Regression
(unit?
unit=22&lesso
n=27)

☐ Statistical
Decision
Theory -
Classification
(unit?
unit=22&lesso
n=28)

☐ Bias -
Variance (unit?
unit=22&lesso
n=29)

☐ **Practice:**
Week 1:
Assignment 1
(Non Graded)
(assessment?
name=264)

☒ Quiz: Week 1:
Assignment 1
(assessment?
name=277)

☒ Week 1
Feedback
Form :
Introduction To
Machine
Learning (unit?
unit=22&lesso
n=189)

☒ Week 1:
Solution (unit?
unit=22&lesso
n=208)

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

Before conditioning dog responds with saliva only in presence of the food but after conditioning it starts salivating just with the bell. Select the correct option(s) about the experiment.

- Presence of just the bell
- Presence of just food
- Presence of both food and bell

- ☐ In this experiment, the dog acts as a Reinforcement learning agent
- ☐ In this experiment, the dog learns in a supervised setting
- ☐ Comparing this experiment to Reinforcement learning theory, the various states are

3) There are n bins of which the k th contains $k - 1$ blue balls and $n - k$ red balls. You **1 point** pick a bin at random and remove two balls at random without replacement. Find the probability that:

- the second ball is red;
- the second ball is red, given that the first is red.

- ☐ 1/3, 2/3
- ☐ 1/2, 1/3
- ☐ 1/2, 2/3
- ☐ 1/3, 1/3

4) A medical company touts its new test for a certain genetic disorder. The false **1 point** negative rate is small: if you have the disorder, the probability that the test returns a positive result is 0.999. The false positive rate is also small: if you do not have the disorder, the probability that the test returns a positive result is only 0.005. Assume that 2% of the population has the disorder. If a person chosen uniformly from the population is tested and the result comes back positive, what is the probability that the person has the disorder?

- ☐ 0.803
- ☐ 0.976
- ☐ 0.02
- ☐ 0.204

5) In an experiment, n coins are tossed, with each one showing up heads with **1 point** probability p independently of the others. Each of the coins which shows up heads is then tossed again. What is the probability of observing 5 heads in the second round of tosses, if we toss 15 coins in the first round and $p = 0.4$?
(Hint: First find the mass function of the number of heads observed in the second round.)

- ☐ 0.372
- ☐ 0.055
- ☐ 0.0345
- ☐ 0.0488

Check Answers and Submit

[Week 6 \(\)](#)[Week 7 \(\)](#)[Week 8 \(\)](#)[Week 9 \(\)](#)[Week 10 \(\)](#)[Week 11 \(\)](#)[Week 12 \(\)](#)[Text
Transcripts
\(\)](#)[Download
Videos \(\)](#)[Books \(\)](#)[Problem
Solving
Session -
July 2024 \(\)](#)